

Question Paper Test 2 (Fall 2025)

Outline of Solution

- (1) (3 points) Let $T : \mathbb{R}^7 \rightarrow \mathbb{R}^5$ be a linear transformation. Suppose $\text{rank}(T) = 3$. Find $\text{nullity}(T)$.

Sol. By the dimension formula, one has

$$\dim(\mathbb{R}^7) - \text{nullity}(T) = \text{rank}(T).$$

Hence $\text{nullity}(T) = 7 - 3 = 4$.

- (2) (5 points) Let

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 2 & 2 \end{pmatrix}$$

Find a basis of the null space $N(A)$.

Sol. We solve the equation $A(x, y, z, w)^T = 0$. By a direct computation, the RREF of A is

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}.$$

Note that y and w are free variables, we see that the solution set, i.e., the null space $N(A)$ is given by

$$\{(-y, y, -w, w)^T : y, w \in \mathbb{R}\}.$$

In particular, we observe that $\{(-1, 1, 0, 0)^T, (0, 0, -1, 1)^T\}$ is a basis for the null space of A .

- (3) (3 points) Let

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

Show that 5 is an eigenvalue of A .

Sol. One way is to compute directly. One can also observe that $A = vv^T$ for $v = (1, 1, 1, 1, 1)^T \in \mathbb{R}^5$. Note that $v^T v = 5$, we see that

$$Av = vv^T(v) = v(v^T v) = 5v.$$

Clearly $v \neq 0$, so 5 is an eigenvalue of A .

- (4) (4 points) Prove or disprove: Let A be an $n \times n$ matrix. Then a scalar $\lambda \in \mathbb{R}$ is an eigenvalue of A if and only if λ is an eigenvalue of A^T .

Proof. It suffices to show the only if direction because $(A^T)^T = A$. Assume $\lambda \in \mathbb{R}$ is an eigenvalue of A . Then $\det(A - \lambda I_n) = 0$. It follows that

$$\det(A^T - \lambda I_n) = \det((A - \lambda I_n)^T) = \det(A - \lambda I_n) = 0.$$

Hence λ is an eigenvalue of A^T . □

- (5) (5 points) Let V be a vector space of dimension n . Let β be an ordered basis for V . Define $T : V \rightarrow \mathbb{R}^n$ given by $T(v) = [3v]_\beta$. Show that T is a linear transformation.

Proof. Let $S : V \rightarrow \mathbb{R}^n$ given by $S(v) := [v]_\beta$ for all $v \in V$. We claim that this is a linear transformation. Write $\beta = \{v_1, \dots, v_n\}$, every $v \in V$ can be uniquely written as $v = \sum_{i=1}^n a_i v_i$ ($a_i \in \mathbb{R}$). Then $S(v) = (a_1, \dots, a_n)^T$.

For any $v = \sum_{i=1}^n a_i v_i$, $w = \sum_{i=1}^n b_i v_i$, we have

$$\begin{aligned} S(v+w) &= [v+w]_\beta = (a_1 + b_1, \dots, a_n + b_n)^T \\ &= (a_1, \dots, a_n)^T + (b_1, \dots, b_n)^T \\ &= [v]_\beta^T + [w]_\beta^T \\ &= S(v) + S(w). \end{aligned}$$

For any $\lambda \in \mathbb{R}$, we also have

$$S(\lambda v) = [\lambda v] = (\lambda a_1, \dots, \lambda a_n)^T = \lambda(a_1, \dots, a_n)^T = \lambda[v]_\beta = \lambda S(v).$$

It follows that S is a linear transformation. It is straightforward to verify that $R : V \rightarrow V$ given by $v \mapsto 3v$ is a linear transformation. Hence the composition

$$\begin{aligned} T : V &\xrightarrow{R} V \xrightarrow{S} \mathbb{R}^n \\ v &\mapsto 3v \mapsto [3v]_\beta \end{aligned}$$

is a linear transformation. □

- (6) (5 points) Let V and W be two vector spaces. Suppose $\dim(V) < \dim(W)$. Show that there exists an injective linear transformation $T : V \rightarrow W$.

Proof. Write $\dim(V) = n$, $\dim(W) = m$, we have $n < m$. Choose a basis $\{v_1, \dots, v_n\}$ of V and a basis $\{w_1, \dots, w_m\}$ of W . Define $T : V \rightarrow W$ via sending v_i to w_i for all $1 \leq i \leq n$. This is a linear transformation by definition. It remains to verify that this is injective. To this end, assume $v = \sum_{i=1}^n a_i v_i \in V$ is such that $T(v) = 0 \in W$. Then $\sum_{i=1}^n a_i w_i = 0$. Since $\{w_i\}_{i=1}^n$ is a linear independent subset of W , one deduces that $a_i = 0$ for all $1 \leq i \leq n$. Hence $v = 0$, i.e., T is injective. □